

DSC Analysis

Distance Sample Count — Wildlife Survey Analysis Pipeline

dsc_analysis

This guide walks you through configuring and running the Distance Sample Count (DSC) Analysis workflow, which ingests wildlife survey patrol data from EarthRanger, calculates distance sampling geometry, labels transects with satellite-derived environmental covariates, and exports analysis-ready datasets for population density modelling.

Overview

The workflow supports **multiple surveys per run**. Each survey is automatically split into its **detected activity periods** — if a survey site was visited in, say, both February and June, the workflow produces two independent, period-tagged output sets instead of one combined one. For each survey period it produces the six files below, plus a matching set of [interactive dashboard](#) views:

Output	Format	Description
<code>{survey}_{period}_analysis_metadata</code>	CSV	Survey event metadata — transect IDs, team members, observer counts, event types, lat/lon
<code>{survey}_{period}_field_effort</code>	CSV	Per-team-member daily distance travelled, duration, and man-hours, derived from EarthRanger GPS tracks
<code>{survey}_{period}_analysis_data</code>	CSV	Wildlife observations enriched with off-transect distances, orthogonal distances, estimated

Output	Format	Description
		animal positions, HLS NDVI, and terrain slope
<code>{survey}_{period}_events</code>	GeoPackage	Spatial point layer of wildlife observations with key distance sampling geometry columns
<code>{survey}_{period}_transect_areas</code>	GeoPackage	Visited transect corridors (buffered) labelled with mean NDVI, terrain slope, and PALSAR woody cover from Google Earth Engine
<code>{survey}_{period}_transect_lines</code>	GeoPackage	Visited transect centrelines (unbuffered), reprojected to EPSG:4326

Prerequisites

Before running the workflow, ensure you have:

- Access to an **EarthRanger** instance with DSC patrol data logged using:
 - `distancecountpatrol_rep` — survey metadata (transect ID, team, observer count)
 - `distancecountwildlife_rep` — wildlife observations (species, count, distance, radial angle)
- A **Google Earth Engine** service account configured in Ecoscope (for NDVI, slope, and woody cover imagery)
- The **EarthRanger spatial group ID** for each survey's transect lines
- The **patrol type ID** (numeric or UUID) for the DSC patrol type in EarthRanger
- Team members recorded on each patrol should correspond to named **subjects/trackers** in EarthRanger, so their GPS tracks can be used to compute field effort

Step-by-Step Configuration

1 Add the Workflow Template

In Ecoscope, go to **Workflow Templates** and click **Add Workflow Template**. Paste the GitHub repository URL into the **Github Link** field and click **Add Template**.

```
https://github.com/wildlife-dynamics/dsc_analysis.git
```

Note: The card may show *Initializing...* briefly while the environment is set up.

2 Select the Workflow

After the template is added it appears in the **Workflow Templates** list as **dsc_analysis**. Click the card to open the workflow configuration form.

3 Set Workflow Details and Time Range

The configuration form opens with two sections at the top.

Set Workflow Details

Field	Description
Workflow Name	A short name to identify this run (required)
Workflow Description	Optional notes, e.g. survey season or region

Time Range

This field is required on all Ecoscope workflows. It now directly controls which patrol events are fetched: patrol events on or after **Since** and up to **Until** are retrieved from EarthRanger for every survey configured in Step 5. It is also used for timestamp display and UTC conversion.

One shared time range covers every survey in the run — there is no longer a separate time window per survey.

Tip: If different surveys need different fetch windows, submit them as separate workflow runs — a single run applies one Time Range to every survey it contains.

Field	Description
Timezone	Local timezone for display and UTC conversion, e.g. <code>Africa/Nairobi (UTC+03:00)</code>
Since	Start of the data fetch window — applied to every survey in this run
Until	End of the data fetch window — applied to every survey in this run

4 Configure Google Earth Engine Connection

Select your Google Earth Engine service account from the **Data Source** dropdown. This connection is used to build HLS NDVI composites and terrain slope images for transect labelling.

Note: Only one GEE connection is needed — it is shared across all surveys in the run.

5 Configure EarthRanger Connection and Survey Details

Scroll down to **Configure EarthRanger and Survey Details**. Click **+ Add** to add a survey entry — each entry is one independent survey, fetched using the shared Time Range from Step 3.

Field	Description
Data Source	Select the EarthRanger connection for this survey (e.g. <code>Amboseli Trust for Elephants</code>)

Field	Description
Patrol Type ID	The numeric or UUID identifier for the DSC patrol type in EarthRanger

Transects Group ID The EarthRanger spatial group ID(s) that contain this survey's transect line features. Click **+ Add** within this field to add more than one group ID if the survey's transects span multiple spatial groups

Multiple surveys: To configure another survey, click the outer **+ Add** button again to add a new entry — whether it points at the same or a different EarthRanger connection, each entry runs as an independent survey.

Running the Workflow

Once all parameters are configured, click **Submit**. For each survey the workflow will:

- 1 Fetch patrol events matching the configured patrol type ID and the workflow's Time Range from EarthRanger.
- 2 Split the survey's patrol events into their detected `{survey}_{yyyy}_{mm}` label. activity periods — each period is processed as an independent branch, tagged with a
- 3 Fetch transect lines from the configured EarthRanger spatial group, once per survey period.
- 4 Retrieve individual wildlife observation events from the patrol event IDs.
- 5 Process and normalise survey metadata events — extract transect ID, team members, and observer count.
- 6 Propagate metadata fields to wildlife observation rows using backward/forward fill within each patrol.
- 7 Compute field effort per team member — daily distance travelled, duration, and man-hours, from EarthRanger subject GPS tracks.

- 8 Reproject events and transects to a UTM coordinate system for metric distance calculations.
- 9 Calculate the perpendicular distance from each observer `off_transect_dist`). position to the transect centreline (
- 10 Estimate the true animal position from each observation's radial `dist_to_centre`). angle and distance to centre (
- 11 Calculate the orthogonal distance from the estimated animal `ortho_dist`). position to the transect centreline (
- 12 Filter events to those that intersect a 500 m transect buffer corridor (rounded end caps); discard unvisited transects — a transect with a patrol "conducted" marker but no wildlife sightings still counts as visited.
- 13 Label visited transects with mean HLS NDVI (max 30 % cloud cover), terrain slope, and PALSAR woody cover (dry-season composite) from Google Earth Engine.
- 14 Merge transect covariates into the wildlife observation dataset.
- 15 Export all outputs per survey period to their type- `$ECOSCOPE_WORKFLOWS_RESULTS` . specific subfolder under

Output Files

Outputs are written under `$ECOSCOPE_WORKFLOWS_RESULTS/` , organized into a subfolder per output type so files are pre-sorted for reviewers instead of landing flat in the results root. Six files are produced for **each survey period** — `{survey}_{period}` is replaced by the name of the EarthRanger site (subdomain) that survey's connection points to, followed by the detected activity period (e.g. `olaremotorogi_2026_02`).



ER_Metadata/{survey}_{period}_analysis_metadata.csv

Survey metadata events: transect IDs, team members, observer counts, event types, lat/lon



ER_FieldEffort/{survey}_{period}_field_effort.csv

Per-team-member daily field effort. Columns: `survey_date` , `id` , `name` , `team_size` , `distance` (km), `duration` (h), `man_hours`



ER_AnalysisData/{survey}_{period}_analysis_data.csv

Wildlife observations with all distance sampling fields: species, total count, juveniles, `dist_to_centre`, `radialangle`, `off_transect_dist`, `ortho_dist`, estimated geometry, `NDVI_HSL`, `slope`, `survey_id`, and more



ER_Events/{survey}_{period}_events.gpkg

Spatial point layer of wildlife observations. Columns: `serial_number`, `transect_id`, `dist_to_centre`, `ortho_dist`, `intersects_transect`, `geometry`



ER_SurveyAreas/{survey}_{period}_transect_areas.gpkg

Visited transect corridors (buffered polygons) in EPSG:4326 with `NDVI_HSL`, `slope`, `WoodyCover`, and `img_date_hsl_ndvi` columns



ER_MasterTransects/{survey}_{period}_transect_lines.gpkg

Visited transect centrelines (unbuffered) in EPSG:4326

Note: If a survey has no visited transects or no matching patrol events for a given period, that period's outputs are skipped gracefully rather than raising an error — other periods and surveys in the same run are unaffected.

Note: A transect counts as "visited" — and so appears in `transect_areas.gpkg` / `transect_lines.gpkg` — if it has a patrol "conducted" marker event, a wildlife sighting, or both; a transect that was walked but had no sightings is no longer dropped from the output. `analysis_data.csv` and `events.gpkg` remain wildlife-sightings-only.

Note: As of this revision, `WoodyCover` is labelled onto the transect corridors and appears in `transect_areas.gpkg`, but is not yet included in the `analysis_data.csv` column selection — only `NDVI_HSL` and `slope` are currently carried through to that file.

Dashboard

In addition to the files above, each run assembles an **interactive dashboard** — one set of views per survey period, selectable through the dashboard's grouper controls (survey, then period):

Widget	Type	Shows
Total Distance Surveyed (km)	Stat tile	Sum of field effort distance for the period
Total Effort (hours)	Stat tile	Sum of field effort duration for the period
Number of Transects Surveyed	Stat tile	Count of distinct transects present in the period
Survey Overview Map	Map	Transect areas (buffered corridor), transect lines, and wildlife event points, auto-framed to the survey's extent
Field Effort	Table	Sortable, filterable, downloadable copy of the field effort data
Survey Metadata	Table	Sortable, filterable, downloadable copy of the analysis metadata
Analysis Data	Table	Sortable, filterable, downloadable copy of the analysis data (geometry columns omitted)

Note: Earlier revisions of this workflow shipped with an empty dashboard — a "files ready to publish" widget was built but never attached to it. All seven widgets above are now wired in and populate automatically on every run.